

Module 1 (Assignment)

Introduction to AI Driven Software Quality Assurance

Q1. What is software Quality Assurance?

- **Define SQA**

Software Quality Assurance (SQA) is systematic processes that ensure software is developed and maintained according to requirements, standards and best practice. It focuses on preventing defects by monitoring and improving the software developing process throughout the Software Development Life Cycle (SDLC).

The primary goal of SQA is to deliver high quality software that is reliable, secure, sufficient and meets customer satisfaction.

Key responsibility of SQA Includes:

1. Defining quality standards and process.
2. Reviewing requirements, design and code.
3. Conducting software Testing.
4. Identifying and reporting defects.
5. Ensuring the final product meets customer and business requirements.

- **Why Quality is important in software developments?**

Quality is essential in software development because it ensures that the software is reliable, functional, secure, and satisfies user needs. High-quality software reduces errors, improves customer satisfaction, and lowers maintenance costs.

Importance of quality in software development:

1. Customer Satisfaction
(High-quality software meets user expectations and provides a positive experience.)
2. Reduce Defects
(Early Identification and prevention of bugs.)
3. Improve Reliability
(Quality software perform consistently without unexpected failure)
4. Enhance Security
(Proper quality assurance helps identify security vulnerabilities before release.)
5. Reduce Maintenance Cost
(Fewer defects after deployment means less time and money spent on fixing issue.)

6. Improve Business Reputation

(Reliable software builds customer trust strengthens the company reputation)

7. Ensure Compliance

(Quality Process help ensure the software follows industry standards regulatory requirements)

Software quality is important because it ensures the software is dependable, efficient, secure, and meets both customer and business requirements. Investing in quality throughout the development process leads to better products, satisfied users, and long-term business success.

Q2. Differentiate between the following:

QA vs QC

Quality Assurance (QA)	Quality Control (QC)
QA is process-oriented.	QC is product-oriented.
It focuses on preventing defects.	It focuses on finding and fixing defects.
QA ensures that the correct development process is followed.	QC checks whether the final product meets quality standards.
QA activities are performed throughout the software development process.	QC activities are mainly performed after the software is developed or during testing.
It includes process reviews, audits, and defining quality standards.	It includes software testing, bug detection, and defect verification.
The goal is to improve the development process.	The goal is to improve the quality of the software product.
Example: Creating coding standards and reviewing development processes.	Example: Executing test cases to identify bugs in an application.

QC vs Testing

Quality Control (QC)	Testing
QC is a quality management process.	Testing is a technique used to find defects in software.
Its goal is to ensure the software meets quality standards.	Its goal is to verify that the software works as expected and identify bugs.
QC includes testing, inspections, reviews, and other quality-checking activities.	Testing is only one part of QC.
QC focuses on the overall quality of the software product.	Testing focuses on checking the functionality, performance, security, usability, etc.
QC is performed by QA engineers, testers, and other quality professionals.	Testing is mainly performed by testers or QA engineers.
Example: Reviewing test results, inspecting deliverables, and ensuring quality standards are met.	Example: Running login test cases to check whether users can log in successfully.

QA vs Testing

Quality Assurance (QA)	Testing
QA is a process-oriented activity.	Testing is a product-oriented activity.
Its main goal is to prevent defects by improving the development process.	Its main goal is to find defects in the software.
QA covers the entire Software Development Life Cycle (SDLC).	Testing is mainly performed during the testing phase of the SDLC.
QA includes process definition, standards, reviews, audits, and continuous improvement.	Testing includes executing test cases, identifying bugs, and verifying fixes.

Quality Assurance (QA)	Testing
QA focuses on how the software is developed.	Testing focuses on whether the software works correctly.
QA is a broader concept.	Testing is a part of QA.
Example: Creating a test strategy, defining coding standards, and conducting process audits.	Example: Testing the login page to verify valid and invalid login scenarios.

Q3: How is Ai changing Software Testing?

Ways AI is changing software testing

1. Automated Test Case Generation
 - AI can automatically create test cases based on software requirements or user behavior, reducing manual effort.
2. Faster Test Execution
 - AI-powered tools can execute thousands of test cases quickly, saving time and increasing productivity.
3. Self-Healing Test Scripts
 - AI can automatically update test scripts when the application's user interface changes, reducing maintenance costs.
4. Defect Prediction
 - AI analyzes historical data to predict which parts of the software are most likely to contain defects.
5. Smart Bug Detection
 - AI helps identify patterns in failures and prioritize the most critical bugs for developers.
6. Regression Testing
 - AI selects the most relevant test cases for regression testing, reducing unnecessary test execution.
7. Visual Testing
 - AI compares screenshots to detect UI issues such as layout changes, missing elements, or incorrect colors.
8. Improved Test Coverage
 - AI identifies untested areas of the application and suggests additional test cases.

Advantages of AI in Software Testing

- Faster testing
- Higher accuracy
- Reduced manual effort
- Early defect detection
- Lower testing costs
- Better test coverage

Limitations of AI

- Requires quality data for accurate results.
- Cannot fully replace human testers.
- Human judgment is still needed for exploratory testing, usability testing, and business decision-making.

AI is changing software testing by automating repetitive tasks, improving accuracy, and speeding up the testing process. However, it works best when combined with human expertise. AI assists testers rather than replacing them, allowing QA teams to focus on complex scenarios and delivering higher-quality software.

Q4. Explain the responsibilities of:

QA Engineer

Responsibilities of QA Engineer:

- Analyze Requirements
 - Review software requirements and understand the expected functionality.
- Create Test Plans and Test Cases
 - Design test plans, test scenarios, and detailed test cases based on the requirements.
- Execute Tests
 - Perform manual or automated testing to verify that the software works correctly.
- Identify and Report Bugs
 - Find defects, document them clearly, and report them to the development team.
- Verify Bug Fixes
 - Retest fixed defects to ensure they have been resolved without creating new issues.
- Perform Different Types of Testing
 - Conduct functional, regression, integration, system, performance, and usability testing as needed.
- Collaborate with the Team
 - Work closely with developers, business analysts, and project managers to ensure software quality.
- Maintain Test Documentation
 - Prepare and update test plans, test cases, test reports, and defect reports.
- Ensure Quality Standards
 - Follow testing standards, processes, and best practices to maintain software quality.
- Support Continuous Improvement
 - Suggest improvements to testing processes and help increase the efficiency of the QA team.

Software Test Engineer:

Responsibility of Software Test Engineer:

1. Understand Software Requirements
 - Analyze the Software Requirements Specification (SRS) and understand the application's functionality.
2. Prepare Test Cases
 - Design and document test scenarios and test cases based on the requirements.
3. Execute Test Cases
 - Perform manual or automated testing to verify that the software works as expected.
4. Identify and Report Defects
 - Detect bugs, document them clearly, and report them using a defect-tracking tool.
5. Verify Bug Fixes
 - Retest resolved defects and perform regression testing to ensure no new issues are introduced.
6. Maintain Test Documentation
 - Keep test cases, test reports, and defect reports up to date.
7. Collaborate with Developers
 - Work closely with developers and other team members to resolve defects and improve software quality.
8. Ensure Software Quality Before Release
 - Confirm that the software meets functional requirements and quality standards before deployment.

AI Quality Engineer:

Responsibility of AI Quality Engineer:

1. Develop AI Models
 - Design and build machine learning and AI models.
2. Prepare and Process Data
 - Collect, clean, and organize data for training AI models.
3. Train and Optimize Models
 - Train AI models and improve their accuracy and performance.
4. Integrate AI into Applications
 - Implement AI models into software applications and systems.
5. Test and Evaluate AI Systems
 - Verify that AI models perform correctly and meet business requirements.
6. Monitor and Maintain AI Models
 - Track model performance and update models when needed.
7. Collaborate with Teams
 - Work with software developers, data scientists, QA engineers, and stakeholders.
8. Document AI Solutions
 - Maintain documentation for AI models, datasets, and development processes.
9. Ensure Security and Ethics

- Develop AI systems that are secure, reliable, fair, and compliant with ethical standards.
10. Research New AI Technologies
- Stay updated with the latest AI tools, frameworks, and techniques.

Q5. Career Roadmap of QA Professionals

